

Northwind Sales Report

MECE Breakdown · Areas · Sub-Areas · KPIs · Data Sources

Overall Performance	Sales Analysis	Customer Analysis	Product Analysis
Orders Analysis	Employee Analysis	Inventory Analysis	Logistics & Shipping

This document provides a structured MECE breakdown of the Northwind analytics framework — mapping each business domain to measurable KPIs and the underlying data source tables required for Power BI report development.

OVERALL PERFORMANCE

Sub-Area	KPIs / Metrics	Data Sources
KPI Summary	Total Revenue, Total Orders, Total Customers, Total Employees, Total Stock	fact_Orders + fact_OrderDetails + dim_Customers + dim_Employees + dim_Products
Revenue Trend	Monthly Revenue Line Chart — performance trajectory over selected period	dim_Date + fact_Orders + fact_OrderDetails
Revenue by Country	Clustered Bar Chart of Revenue by Customer Country/Region	dim_Customers + fact_Orders
Customer Type Mix	Donut Chart — New vs Returning/Loyal Customer Split	dim_Customers + fact_Orders
Shipping Performance	Avg Days to Ship by Carrier — Logistics Health Indicator	dim_Shippers + fact_Orders
Interactive Filters	Slicers for Date, Country, Region, Shipper, Employee — dynamic cross-filtering	dim_Date + dim_Customers + dim_Shippers + dim_Employees

SALES ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Sales Trends	Revenue MoM %, Revenue YoY %, YTD vs LY, Monthly & Quarterly Trends	dim_Date + fact_Orders + fact_OrderDetails
Revenue Performance	Total Revenue, Total Orders, Avg Order Value	fact_OrderDetails
Regional Sales	Revenue by Country/Region, AOV by Country, Top Markets	dim_Customers + fact_Orders
Discount Impact	Discount Amount Lost, Avg Discount %, Before vs After Revenue	fact_OrderDetails

CUSTOMER ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Customer Segmentation	Active Customers, RFM Segments, New vs Returning	dim_Customers + fact_Orders
Customer Value	Revenue per Customer, Top 10% Contribution, CLV	dim_Customers + fact_Orders
Customer Loyalty	Retention Rate, Churn Indicators, Loyal Customers	fact_Orders + dim_Customers
Purchase Behavior	Repeat Rate, Avg Orders per Customer, Days Between Orders	fact_Orders
Geographic Distribution	Customer Count by Region, Revenue by Country	dim_Customers

PRODUCT ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Product Performance	Top Products, Order Frequency, Pareto (Top 20%), Best/Worst Sellers	dim_Products + fact_OrderDetails
Category Analysis	Revenue by Category, Mix %, Units Sold, Growth Trend	dim_Categories + fact_OrderDetails
Pricing Analysis	Avg Price, Price Bands, Price Impact on Revenue	dim_Products
Product Lifecycle	Discontinued Impact, Product Stages, Phase-out Candidates	dim_Products + fact_OrderDetails

ORDERS ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Order Volume Trends	Total Orders Over Time, MoM/YoY Orders, Quarterly Orders	dim_Date + fact_Orders
Order Frequency	Orders per Day/Week/Month, Avg Orders per Customer, Distribution	fact_Orders
Peak Period Analysis	Seasonal Peaks, Busy Periods, Holiday Impact	dim_Date + fact_Orders
Demand Patterns	Seasonality Index, Demand Consistency, Stability	dim_Date + fact_Orders

EMPLOYEE ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Sales per Employee	Revenue per Employee, Orders per Employee, AOV	dim_Employees + fact_Orders
Orders per Employee	Monthly Trends, Ranking, Distribution of Orders	fact_Orders
Regional Contribution	Revenue by Region per Employee, Coverage Area, Market Dominance	dim_Employees + dim_Customers + fact_Orders
Performance Ranking	Revenue vs Tenure, Team Roll-up, Performance Score	dim_Employees
Performance Contribution	Revenue Contribution %, Top Performers, Bottom Performers	dim_Employees + fact_Orders + fact_OrderDetails

INVENTORY ANALYSIS

Sub-Area	KPIs / Metrics	Data Sources
Inventory Status	Units in Stock, Units on Order, Active Products	dim_Products
Stock Health	Products Below Reorder Level, Stock-Out Risk Count	dim_Products
Inventory Distribution	Stock by Category, Stock by Supplier	dim_Products + dim_Suppliers
Inventory Efficiency	Approx Inventory Turnover Ratio, Slow-moving Products	fact_OrderDetails + dim_Products
Demand vs Supply	High Demand + Low Stock Alerts, Overstock Risk Products	fact_OrderDetails + dim_Products

LOGISTICS & SHIPPING

Sub-Area	KPIs / Metrics	Data Sources
Shipping Time Analysis	Avg Days to Ship, On-Time Delivery %, Speed Trend Over Time	dim_Shippers + fact_Orders
Freight Cost Analysis	Freight per Order, Total Freight Cost, Freight as % of Revenue	fact_Orders
Shipper Comparison	Carrier Performance, Best/Worst Carrier, Cost vs Speed	dim_Shippers + fact_Orders
Delivery Delays	Late Orders %, Avg Delay Days, Total Delay Count	fact_Orders